In [6]:
```
t = ()
t
```
Out[6]:
```
()
```
In [7]:
```
type(t)
```
Out[7]:
```
tuple
```
In [8]:
```
t1 = (10, 20, 20)
```
In [9]:
```
t1
```
Out[9]:
```
(10, 20, 20)
```
In [10]:
```
t2 = (10, 2.3, 'nit', True, 1+2j)
```
In [11]:
```
t2
```
Out[11]:
```
(10, 2.3, 'nit', True, (1+2j))
```
In [12]:
```
print(t)
print(t1)
print(t2)
```
```
()
(10, 20, 20)
(10, 2.3, 'nit', True, (1+2j))
```
In [13]:
```
t1.count(10)
```
Out[13]:
```
1
```
In [14]:
```
t1
```
Out[14]:
```
(10, 20, 20)
```
In [15]:
```
t1.index(20)
```
Out[15]:
```
1
```
In [16]:
```
t1
```
Out[16]:
```
(10, 20, 20)
```
In [17]:
```
t1.append(10)
```
```
---------------------------------------------------------------------------
AttributeError                            Traceback (most recent call last)
Cell In[17], line 1
----> 1 t1.append(10)

AttributeError: 'tuple' object has no attribute 'append'
```
In [18]:
```
t1[0]
```
Out[18]:
```
10
```
In [20]:
```
t1[0] = 100
```
```
  Cell In[20], line 1
    t1(0) = 100
    ^
SyntaxError: cannot assign to function call here. Maybe you meant '==' instead of '='?
```
In [22]:
```
t1
```
Out[22]:
```
(10, 20, 20)
```
In [23]:
```
t1.pop()
```
```
---------------------------------------------------------------------------
AttributeError                            Traceback (most recent call last)
Cell In[23], line 1
----> 1 t1.pop()

AttributeError: 'tuple' object has no attribute 'pop'
```
In [24]:
```
t1
```

Out[24]: (10, 20, 20)

In [25]:
```python
t1.remove(20)
```

```
---------------------------------------------------------------------------
AttributeError                            Traceback (most recent call last)
Cell In[25], line 1
----> 1 t1.remove(20)

AttributeError: 'tuple' object has no attribute 'remove'
```

In [26]:
```python
t1
```

Out[26]: (10, 20, 20)

In [27]:
```python
t1 * 3
```

Out[27]: (10, 20, 20, 10, 20, 20, 10, 20, 20)

In [28]:
```python
s = {}
s
```

Out[28]: {}

In [29]:
```python
type(s)
```

Out[29]: dict

In [30]:
```python
s1 = {100, 4, 10, 33, 87, 20, 30}
```

In [31]:
```python
s1
```

Out[31]: {4, 10, 20, 30, 33, 87, 100}

In [32]:
```python
type(s1)
```

Out[32]: set

In [33]:
```python
s2 = {3, 1+2j, 4.5, True}
```

In [34]:
```python
s2
```

Out[34]: {(1+2j), 3, 4.5, True}

In [35]:
```python
print(s)
print(s1)
print(s2)
```

```
{}
{33, 100, 4, 20, 87, 10, 30}
{True, 3, 4.5, (1+2j)}
```

In [37]:
```python
print(type(s))
print(type(s1))
print(type(s2))
```

```
<class 'dict'>
<class 'set'>
<class 'set'>
```

In [38]:
```python
s1
```

Out[38]: {4, 10, 20, 30, 33, 87, 100}

In [39]:
```python
s1.add(25)
```

In [40]:
```python
s1
```

Out[40]: {4, 10, 20, 25, 30, 33, 87, 100}

In [41]:
```python
s3 = s1.copy()
```

In [42]:
```python
s3
```

Out[42]: {4, 10, 20, 25, 30, 33, 87, 100}

In [43]:
```python
print(s)
print(s2)
```

```
print(s3)
```

```
{}
{True, 3, 4.5, (1+2j)}
{33, 100, 4, 10, 20, 87, 25, 30}
```

In [44]:
```
s3.clear()
```

In [45]:
```
print(s)
print(s2)
print(s3)
```

```
{}
{True, 3, 4.5, (1+2j)}
set()
```

In [56]:
```
s1.pop()
```

Out[56]:
```
87
```

In [57]:
```
s1
```

Out[57]:
```
{10, 25, 30}
```

In [58]:
```
s1.discard(1000)
```

In [59]:
```
s1.remove(1000)
```

```
---------------------------------------------------------------------------
KeyError                                  Traceback (most recent call last)
Cell In[59], line 1
----> 1 s1.remove(1000)

KeyError: 1000
```

In [60]:
```
s1
```

Out[60]:
```
{10, 25, 30}
```

In [61]:
```
s1.discard(10)
```

In [62]:
```
s1
```

Out[62]:
```
{25, 30}
```

In [63]:
```
s1.add(30)
```

In [64]:
```
s1
```

Out[64]:
```
{25, 30}
```

In [ ]:
```

```